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ChE 210 Quiz #1 13 September 2019

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In class, we discussed fitting human blood viscosity data using the Sisko Model for a shear-rate-dependent viscosity $\eta(\dot{\gamma})$: $\eta - \eta_\infty = m\,\dot{\gamma}^{n-1}$
where η_∞, m, and n are three adjustable constants. It is know that for human blood that $n = 0.5$ is a good approximation.

(a) Assuming that $n = 0.5$, how can the Sisko model be linearized so that linear least squares can be used to determine the remaining two constants? Show clearly what you would plot as x on the horizontal axis and y on the vertical axis.

(b) Use the data provided to calculate the remaining two parameters using horizontal offsets. Be sure to give the correct units.

For horizontal offsets: slope $a = \frac{S_{yy}-S_y^2}{S_{xy}-S_xS_y}$ and intercept $b = \frac{S_{xy}S_y-S_{yy}S_x}{S_{xy}-S_xS_y}$.

Shear Rate $\dot{\gamma}$ (s^{-1})	Viscosity η (mPa·s)
0.51	39.15
1.34	22.18
7.97	9.51
90.73	5.77
759.07	4.23

$S_x = 860$

$S_y = 16.2$

a) $\underbrace{\eta}_{y} = \underbrace{m}_{a}\underbrace{\dot{\gamma}^{-0.5}}_{x} + \underbrace{\eta_{\infty}}_{b}$ YES!

b) $a = \frac{433-262}{772-13932} = -0.0\cancel{123}$

$b = \frac{12506-372380}{772-13932} = 2\cancel{7.}346$

Show your work!